Level 3
Advanced Machine Learning & Natural Language Processing

Customer Churn Prediction

Objective

The objective of this task was to build classification models to predict customer churn using structured telecom data. Two models were implemented:

- Logistic Regression
- Random Forest

The goal was to compare predictive performance and identify the most important churn drivers.

Data Quality Assessment

- No missing values detected
- No duplicate observations found
- Target variable converted to factor
- Categorical predictors properly encoded

The dataset was clean and suitable for modeling.

Logistic Regression Results

Confusion Matrix

	Actual No	Actual Yes
Predicted No	549	71
Predicted Yes	23	24

Performance Metrics

- Accuracy: 85.91%
- Precision: 51.06%
- Recall: 25.26%
- F1 Score: 0.338

Interpretation

The logistic regression model achieved good overall accuracy. However:

- Recall is low (25%), meaning many churners were missed.
- The model struggles to correctly identify customers who will churn.
- Class imbalance likely affects performance.

Key Significant Predictors (Logistic Model)
Highly significant variables include:

- International Plan (Strong positive impact)
- Customer Service Calls (Strong positive impact)
- Total International Calls
- Voice Mail Plan (Negative impact)

Insight
Customers with an international plan and high customer service calls are significantly more likely to churn.

Random Forest Results
Confusion Matrix

	Actual No	Actual Yes
Predicted No	567	28
Predicted Yes	5	67

Performance Metrics

- Accuracy: 95.05%
- Precision: 93.06%
- Recall: 70.53%
- F1 Score: 0.802

Interpretation
The Random Forest model significantly outperforms Logistic Regression:

- Much higher recall (70%)

- Much better F1 score
- Strong ability to detect churners

This demonstrates the advantage of ensemble methods for complex classification tasks.

Feature Importance (Random Forest)
Top influential variables:

- Customer Service Calls
- International Plan
- Total Day Minutes
- Total Day Charge
- Total International Calls

Business Insight
High customer service complaints and high international usage are strong churn indicators. Telecom companies should target these customers for retention strategies.

Sentiment Analysis (Natural Language Processing)
1. Objective
The goal was to classify text data into:

- Positive
- Negative
- Neutral

Using lexicon-based sentiment scoring (Bing lexicon).

2. Text Preprocessing Steps

- Lowercasing text
- Removing punctuation and numbers
- Removing stopwords
- Stemming words
- Tokenization

- Lexicon matching

This ensures clean and structured textual data.

Predicted Sentiment Distribution
From your bar chart:

- Positive sentiments dominate
- Negative sentiments are moderate
- Neutral sentiments are lowest

This suggests the dataset contains generally positive opinions.

Word Cloud Interpretation
The most frequent words include:

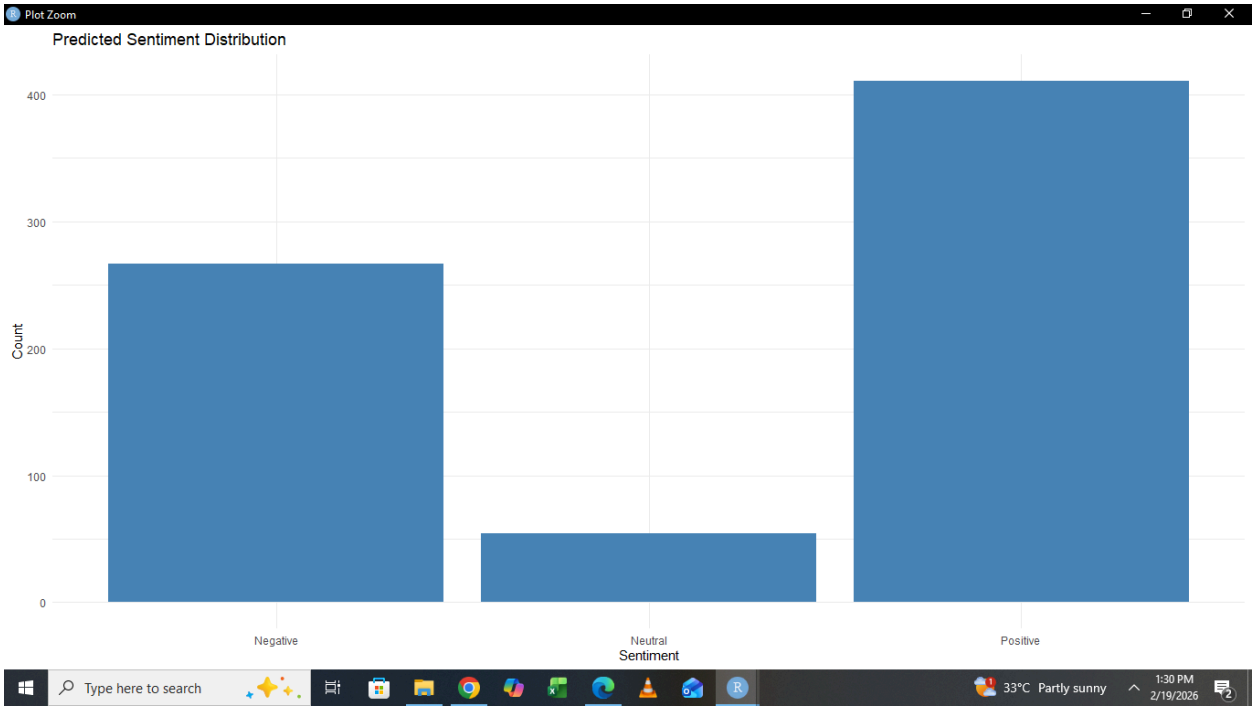
- "new"
- "life"
- "dream"
- "joy"
- "love"
- "friend"
- "challenge"

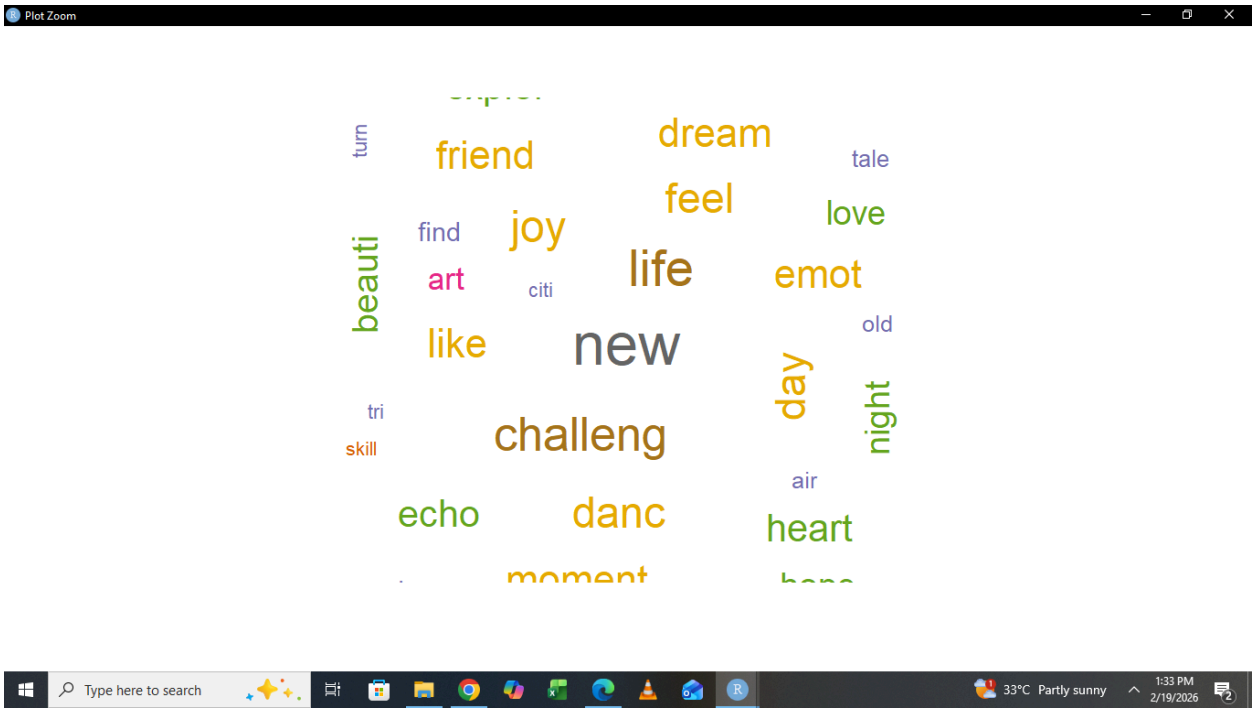
These words reinforce the positive emotional tone detected in the dataset.

Model Observations
Some mismatches between actual and predicted sentiment were observed. This is expected because:

- Lexicon-based models do not understand context
- Sarcasm and emoji may affect classification
- Neutral detection is limited in rule-based system

Visualisations


Plot Zoom
Predicted Sentiment Distribution
Count
400
300
200
100
0
Negative
Neutral
Positive
Sentiment
Type here to search
33°C Partly sunny
1:30 PM
2/19/2026


Plot Zoom
turn
friend
dream
tale
feel
love
find
joy
beauti
art
citi
life
emot
old
like
new
day
night
tri
skill
challeng
air
echo
danc
heart
Type here to search
33°C Partly sunny
1:33 PM
2/19/2026